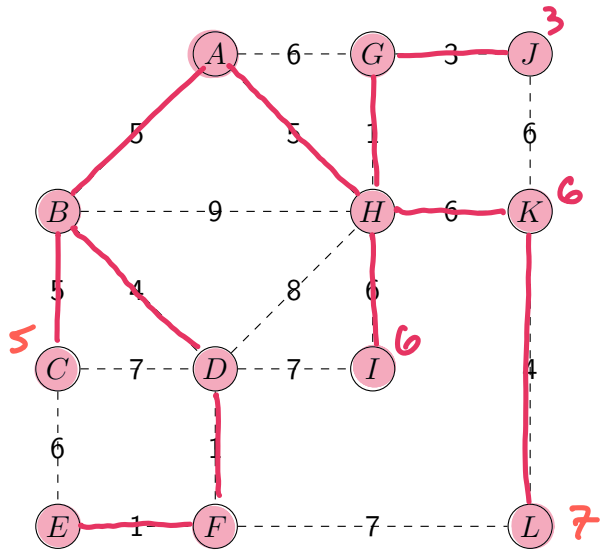
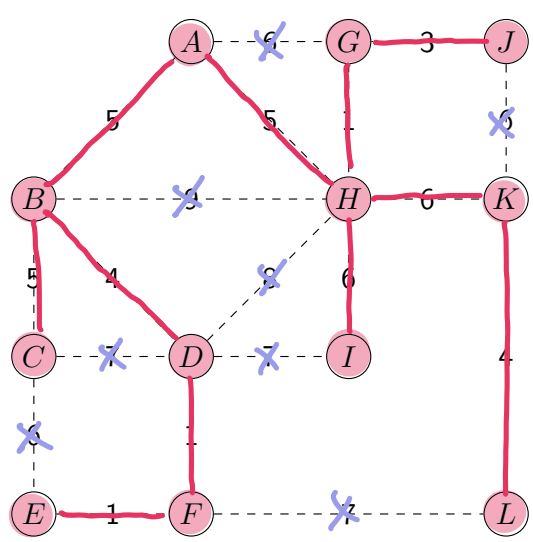
**Problem 1.** Use **Prim's Algorithm** starting at vertex $A$ to determine the minimum spanning tree of the undirected weighted graph given below. **List the order in which the vertices are added** to the minimum spanning tree and **draw the minimum spanning tree** on the graph.




ABDFEHGJCIKL

**Problem 2.** Use **Kruskal's Algorithm** to determine the minimum spanning tree of the undirected weighted graph given below. **List the order in which the edges are added** to the minimum spanning tree and **draw the minimum spanning tree** on the graph.




GH
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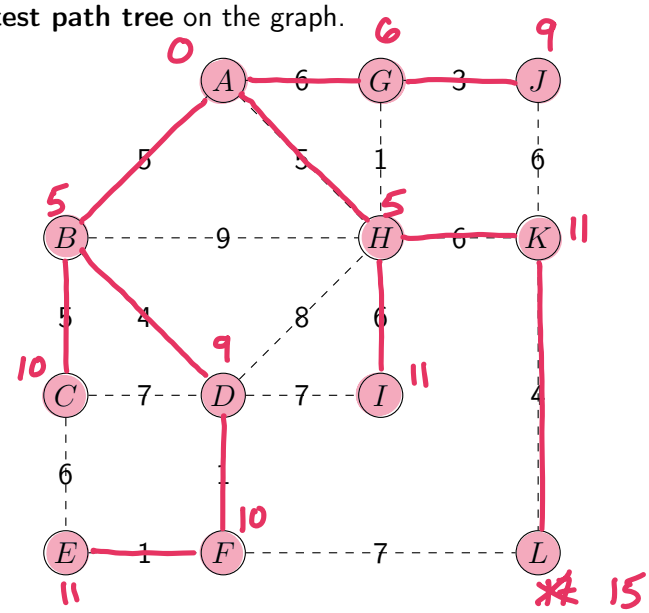
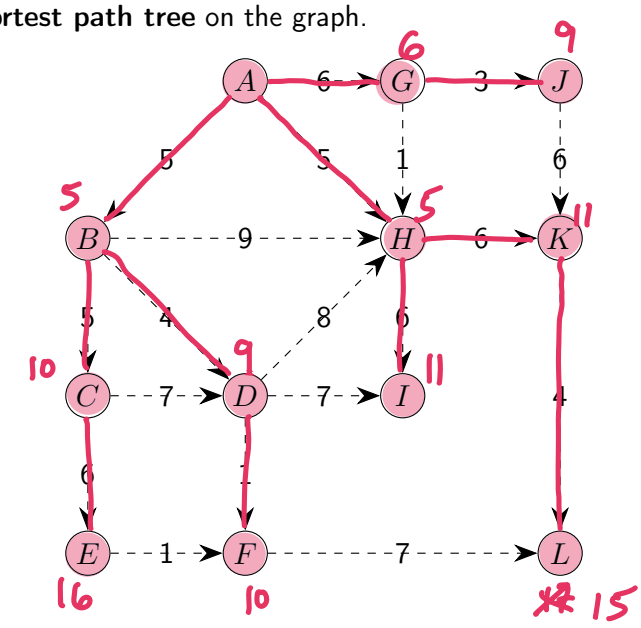
BC
HK
HI

ONLY THE SAME BY CHANCE.

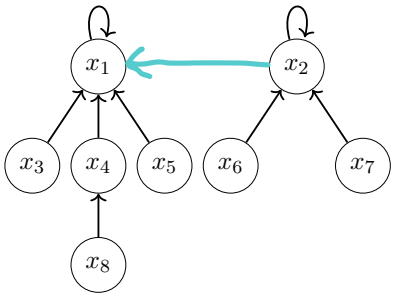
**Problem 3.** Use **Dijkstra's Algorithm** starting at vertex $A$ to determine the shortest path from $A$ to every other vertex in the undirected weighted graph given below. **List the distance** from $A$ to every other vertex on the graph and **draw the shortest path tree** on the graph.




**Problem 4.** Use **Dijkstra's Algorithm** starting at vertex $A$ to determine the shortest path from $A$ to every other vertex in the undirected weighted graph given below. **List the distance** from $A$ to every other vertex on the graph and **draw the shortest path tree** on the graph.



**Problem 5.** The following trees represent two disjoint sets,




Perform the following operations. Be sure to write what is returned by each `FindSet` operation.

1: `UnionByHeight`($x_1$,$x_7$)

2: `FindSet`($x_4$) $x_1$

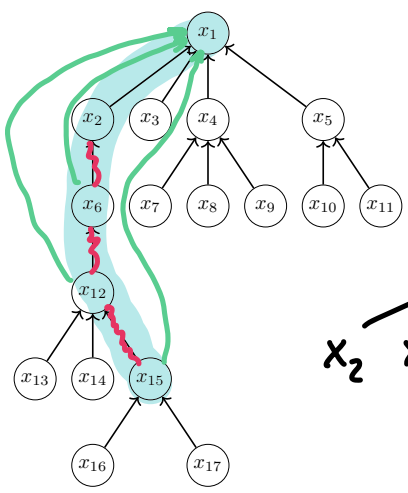
**Problem 6.** The following linked lists represent two disjoint sets,

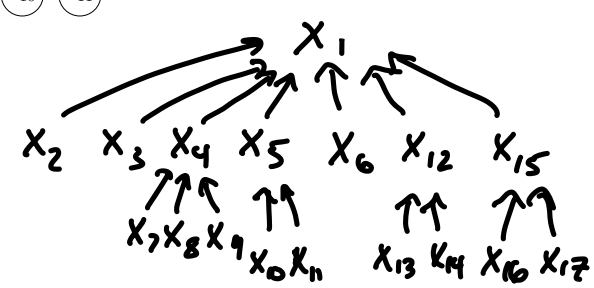
Perform the following operations. Be sure to write what is returned by each `FindSet` operation.

1: `WeightedUnion`($x_3$,$x_5$)

2: `FindSet`($x_6$) $x_5$

**Problem 7.** The following tree represents a set within a Union-Find data structure,




Run the `FindSetWithPathCompression` operation on $x_{15}$.

**Problem 8.** (a) Prove the **Minimum Spanning Forest Problem** is in **P**.

**Minimum Spanning Forest Problem:** Given a undirected weighted graph $G = (V, E)$ and integer $K$, does $G$ have a spanning forest with total weight $\leq K$?

Given $G, K$

Run Kruskal's Alg. on $G$. ← poly. time

Check if weight(all trees) $\leq K$ ← $O(n+m)$

Thus, solveable in poly. time

So, $\in P$.

**Definition 1.** A **Graceful Labeling** of a graph with $m$ edges is a labeling of its vertices with some subset of the integers from $0$ to $m$ inclusive, such that no two vertices share a label, and each edge is *uniquely* identified by the absolute difference between its endpoints, such that this magnitude lies between $1$ and $m$ inclusive.[1]

(b) Prove that the **Graceful Labeling Problem** is in **NP**.

**Graceful Labeling Problem:** Given a graph $G = (V, E)$, can $G$ be gracefully labeled?

ex:

Given $G$ with labeled nodes.

1. Check all vertex labels are unique $O(n^2)$
2. Check all vertex labels $\in [0, m]$ $O(n^2)$
3. Compute edge weights // optional. $O(m)$
4. Check all edge weights are unique $O(m^2)$
5. Check all edge weights $\in [1, m]$ $O(m^2)$

So, can VERIFY in poly time. Thus, $\in NP$

[1] Definition from Wikipedia

```
usedLabels = ∅
foreach vertex v in G.V :
    if v.label ∈ usedLabels OR v.label ∉ [0, m]
        return false
    else
        Add v.label to usedLabels
    endif
endfor
Compute edge weights from labels. // optional
usedEdges = ∅
foreach edge (u,v) ∈ G.E
    if w(u,v) ∈ usedEdges OR w(u,v) ∉ [1,m]
        return false
    else
        add w(u,v) to usedEdges.
    endif
endfor
```

$O(n^2)$

$O(M^2)$

$O(n^2+m^2)$